

Software Project Management

Smart Library Manager (مكتبتي الذكية)

Complete Project Management Documentation

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1. Feasibility Study

This feasibility study evaluates whether the proposed Smart Library Manager project is practical, achievable, and worth pursuing given the available time (3-4 weeks), team size (3 members), and technical skill set of the team.

2. Technical Feasibility

The project relies entirely on modern, well-documented, and free/open-source web technologies: React, TypeScript, and Tailwind CSS for the frontend, the free public Open Library API for book metadata retrieval, and the browser's built-in LocalStorage for persistent data storage. All required tools (Node.js, npm, VS Code, Git/GitHub) are freely available and already familiar to the team from coursework.

No paid infrastructure, servers, or databases are required, since the application runs fully client-side in the browser. This significantly reduces technical risk and setup overhead.

Conclusion: The project is technically feasible.

3. Economic Feasibility

Since the project uses only free and open-source tools and a free public API, there is no direct monetary cost involved in development. The only 'cost' is the time and effort invested by the three team members, which is estimated in the Cost Estimation document.

Conclusion: The project is economically feasible, as it requires no financial investment beyond student time and effort.

4. Operational Feasibility

The application directly addresses a real, common need: helping an individual reader organize, track, and search their personal book collection. The core operations (adding, editing, deleting, searching, and tracking reading status) are simple and intuitive for end users, requiring no training.

Conclusion: The project is operationally feasible.

5. Schedule Feasibility

Based on the Work Breakdown Structure and Activity List developed for this project, the total estimated duration is 21 working days (approximately 4 calendar weeks), which fits within the timeframe available to the team (3-4 weeks). The Critical Path Method analysis confirms this timeline is achievable given the parallel work capacity of a 3-person team.

Conclusion: The project is feasible within the available schedule.

6. Overall Conclusion

Based on the technical, economic, operational, and schedule feasibility analysis above, the Smart Library Manager project is confirmed to be FEASIBLE and is recommended to proceed to the planning and execution phases.

2. Project Charter

Smart Library Manager (مكتبتي الذكية)

2. Project Purpose / Justification

Many readers struggle to keep track of the books they own, are currently reading, have finished, or wish to read in the future, especially across physical and digital collections. The Smart Library Manager addresses this problem by providing a single, easy-to-use web application to organize, search, and track a personal book collection.

3. Project Objectives

- Deliver a fully functional personal library management web application within 3-4 weeks.
- Enable full CRUD (Create, Read, Update, Delete) management of books and their reading status.
- Integrate the Open Library API to allow smart search and automatic import of book details and covers.
- Provide an interactive statistics dashboard and a full activity log with export capability.
- Ensure data persistence using LocalStorage so no data is lost between sessions.
- Produce complete project management documentation as part of the course requirements.

4. Project Scope (Summary)

In scope: book cataloguing and CRUD operations, dual search (local + Open Library API), statistics dashboard, activity log with CSV/JSON export, card/table view with filtering, keyboard shortcuts, LocalStorage persistence, and an Arabic educational section explaining the tech stack.

Out of scope: multi-user accounts, cloud database/server backend, mobile native applications, and payment/subscription features.

5. Key Stakeholders

Stakeholder	Role / Interest
Project Team	Design, develop, test, and document the application
Course Instructor	Evaluates the project against academic and PM requirements
End Users (Readers)	Use the application to manage their personal book collections
Open Library API	External data provider for book metadata and covers

6. Project Team & Roles

Name	ID	Primary Role
Omnia Muhammed	24030099	Project Lead / Frontend Development (CRUD & Dashboard)
Menna Mahmoud	24030153	Frontend Development (Search, API Integration) / QA
Haneen Ibrahim	24030152	UI/UX Design, Documentation & Project Management

7. High-Level Timeline

Estimated total duration: 21 working days (~4 weeks), covering initiation & planning, design, development, testing, and documentation/deployment phases. See the Gantt Chart and Activity List documents for full details.

8. Success Criteria

- All core features listed in the scope are implemented and working without critical bugs.
- The application is delivered on time, within the 3-4 week schedule.
- All project data persists correctly via LocalStorage across browser sessions.
- Complete and consistent project management documentation is submitted.

9. Authorization

This charter authorizes the project team to proceed with planning and execution of the Smart Library Manager project as part of the Software Project Management course requirements.

3. Project Scope Statement

The Smart Library Manager is a client-side web application that allows a user to organize, search, and track their personal book collection, integrating with the free Open Library API for automatic book data import, and persisting all data locally in the browser.

2. Project Deliverables

- A working web application built with React, TypeScript, and Tailwind CSS.
- Book management module (Create, Read, Update, Delete + reading status tracking: Reading, Finished, Available, Wishlist).
- Dual search feature: local library search + Open Library API search with one-click import.
- Interactive statistics dashboard (Bento Grid layout) and full activity log with CSV/JSON export and print support.
- Card view and table view with category/status filtering, plus keyboard shortcuts (Ctrl+K search, Ctrl+B add).
- Persistent local storage of all application data.
- An Arabic educational section explaining how the application was built.
- Complete project management documentation set (this document and the accompanying PM artifacts).

3. Acceptance Criteria

#	Criterion
1	User can add, edit, delete, and update the status of a book without errors
2	Local search returns matching books instantly; API search returns results from Open Library
3	One-click import correctly adds book title, author, and cover image
4	Dashboard statistics accurately reflect the current state of the library
5	Activity log records every add/edit/delete action and can be exported as CSV/JSON
6	All data remains intact after closing and reopening the browser
7	Keyboard shortcuts (Ctrl+K, Ctrl+B) function as expected

4. Project Exclusions

- No user authentication or multi-user account system.
- No server-side database or cloud backend (data is stored locally in the browser only).
- No native mobile application (web-based/responsive only).
- No payment, subscription, or e-commerce functionality.

5. Constraints

- Fixed timeline of 3-4 weeks (academic deadline).
- Team of only 3 members with shared academic workload from other courses.
- Dependency on the availability and rate limits of the free Open Library API.
- No budget for paid tools, hosting, or third-party services.

6. Assumptions

- All team members have basic working knowledge of React and TypeScript, or can learn it quickly.
- The Open Library API will remain publicly accessible and free throughout development.
- Target users access the application through a modern desktop or laptop web browser.
- The browser's LocalStorage capacity is sufficient for a personal book collection.

4. Work Breakdown Structure (WBS)

The WBS decomposes the Smart Library Manager project into five main deliverable-based phases, each broken down into manageable work packages.

WBS Code	Work Package	Level
1.0	Project Management	Phase
1.1	Project Initiation & Requirements Gathering	Task
1.2	Feasibility Study & Planning	Task
1.3	Monitoring, Control & Closure	Task
2.0	Requirements & Design	Phase
2.1	UI/UX Design (Wireframes)	Task
2.2	System Architecture & Data Model Design	Task
3.0	Development	Phase
3.1	Environment Setup (React / TypeScript / Tailwind)	Task
3.2	Book CRUD Module Development	Task
3.3	Search & Open Library API Integration	Task
3.4	Dashboard & Statistics (Bento Grid)	Task
3.5	Activity Log & Export (CSV / JSON)	Task
3.6	View Modes, Filtering & Keyboard Shortcuts	Task
3.7	LocalStorage Persistence Integration	Task
3.8	Arabic Educational Section	Task
4.0	Testing	Phase
4.1	Unit & Integration Testing	Task
4.2	Bug Fixing & User Acceptance Testing	Task
5.0	Deployment & Documentation	Phase
5.1	GitHub Repository Setup & Upload	Task
5.2	Final Report & Presentation Preparation	Task

5. Activity List

The following table lists every activity derived from the WBS, its estimated duration in working days, and its logical predecessor(s), used as the basis for the Network Diagram, CPM, and Gantt Chart.

ID	Activity	Duration (days)	Predecessor(s)
A	Project Initiation & Requirements Gathering	2	-
B	Feasibility Study & Planning	2	A
C	UI/UX Design (Wireframes)	3	B
D	System Architecture & Data Model Design	2	B
E	Environment Setup (React/TS/Tailwind)	1	D
F	Book CRUD Module Development	4	C, E
G	Search & Open Library API Integration	3	E
H	Dashboard & Statistics (Bento Grid)	3	F
I	Activity Log & Export (CSV/JSON)	2	F
J	View Modes, Filtering & Shortcuts	2	F
K	LocalStorage Persistence Integration	1	F, G
L	Arabic Educational Section	2	K
M	Testing & Bug Fixing	3	H, I, J, L
N	Documentation & GitHub Repository Setup	2	M
O	Final Report & Presentation Preparation	2	N

Total estimated project duration (via Critical Path Method): 21 working days (~4 calendar weeks).

6. Cost Estimation

Since this is an academic project using only free and open-source tools (React, TypeScript, Tailwind CSS) and a free public API (Open Library), there is no real monetary budget. To practice standard project cost estimation techniques, a notional academic labor rate of \$12 per person-day is used to represent the value of team effort (Analogous/Parametric estimation based on estimated effort per WBS phase).

2. Cost Estimate by WBS Phase

WBS Phase	Estimated Effort (person-days)	Rate (\$/person-day)	Estimated Cost (\$)
Project Management	6	12	72
Requirements & Design	8	12	96
Development	21	12	252
Testing	6	12	72
Deployment & Documentation	4	12	48
Subtotal (Labor)	45	-	540

3. Additional Cost Items

Item	Cost (\$)	Notes
Development Tools (VS Code, Node.js, Git)	0	Free / open-source
Open Library API	0	Free public API, no usage fee
Hosting (GitHub Pages, optional)	0	Free tier
Contingency Reserve (10%)	54	Reserve for unforeseen risks/delays

4. Total Estimated Project Cost (Budget at Completion - BAC)

Total Estimated Cost (BAC) = \$540 (labor) + \$54 (10% contingency) = \$594

This notional BAC value is used as the baseline for the Earned Value Management (EVM) analysis in the following document.

7. Earned Value Management (EVM)

Metric	Value
Budget at Completion (BAC)	\$594
Planned Duration	21 working days
Status Date #1 (Mid-project check)	End of Week 3 (Day 14)
Status Date #2 (Project completion)	Day 21

2. EVM Analysis at Mid-Project Checkpoint (Day 14)

Metric	Formula	Value
Planned Value (PV)	$(14/21) \times \text{BAC}$	\$396
Earned Value (EV)	% work completed x BAC (60% actually done)	\$356
Actual Cost (AC)	Recorded team effort cost to date	\$380
Cost Variance (CV = EV - AC)	-	\$-24
Schedule Variance (SV = EV - PV)	-	\$-40
Cost Performance Index (CPI = EV/AC)	-	0.94
Schedule Performance Index (SPI = EV/PV)	-	0.90

Interpretation: At the mid-project checkpoint, CPI = 0.94 (< 1) and SPI = 0.90 (< 1), indicating the project was slightly OVER budget and slightly BEHIND schedule at this point. This was mainly due to the Open Library API integration (Activity G) taking longer than planned. The team responded by reallocating a member temporarily to speed up development.

3. EVM Analysis at Project Completion (Day 21)

Metric	Formula	Value
Earned Value (EV)	100% complete	\$594
Actual Cost (AC)	Total actual effort cost	\$615
Cost Variance (CV = EV - AC)	-	\$-21
Schedule Variance (SV = EV - PV, PV=BAC at completion)	-	\$0 (on time)
Cost Performance Index (CPI = EV/AC)	-	0.97

Interpretation: By the end of the project, the team delivered all planned scope on time (SV = 0), but at a slightly higher cost than budgeted (CPI = 0.97), reflecting the extra effort spent recovering the schedule slippage identified at the mid-project checkpoint. This is a common and acceptable trade-off in short, fixed-deadline academic projects.

8. PERT Analysis

For each activity, Optimistic (O), Most Likely (M), and Pessimistic (P) duration estimates (in working days) were collected, and the PERT Expected Time (TE) and Variance were calculated using the standard formulas:

$$TE = (O + 4M + P) / 6 \quad \text{Variance} = ((P - O) / 6)^2$$

ID	Optimistic (O)	Most Likely (M)	Pessimistic (P)	Expected Time (TE)	Variance
A	1	2	3	2.00	0.111
B	1	2	3	2.00	0.111
C	2	3	5	3.17	0.250
D	1	2	4	2.17	0.250
E	1	1	2	1.17	0.028
F	3	4	6	4.17	0.250
G	2	3	5	3.17	0.250
H	2	3	5	3.17	0.250
I	1	2	4	2.17	0.250
J	1	2	4	2.17	0.250
K	1	1	2	1.17	0.028
L	1	2	3	2.00	0.111
M	2	3	5	3.17	0.250
N	1	2	3	2.00	0.111
O	1	2	4	2.17	0.250

(Highlighted rows indicate activities on the project's critical path.)

2. Expected Project Duration

Sum of TE along the critical path (A-B-C-F-H-M-N-O): 21.85 working days ($\approx$ 22 working days).

Sum of Variance along the critical path: 1.583

Project Standard Deviation (SD) = $\sqrt{1.583} \approx 1.26$ working days.

3. Probability of On-Time Completion

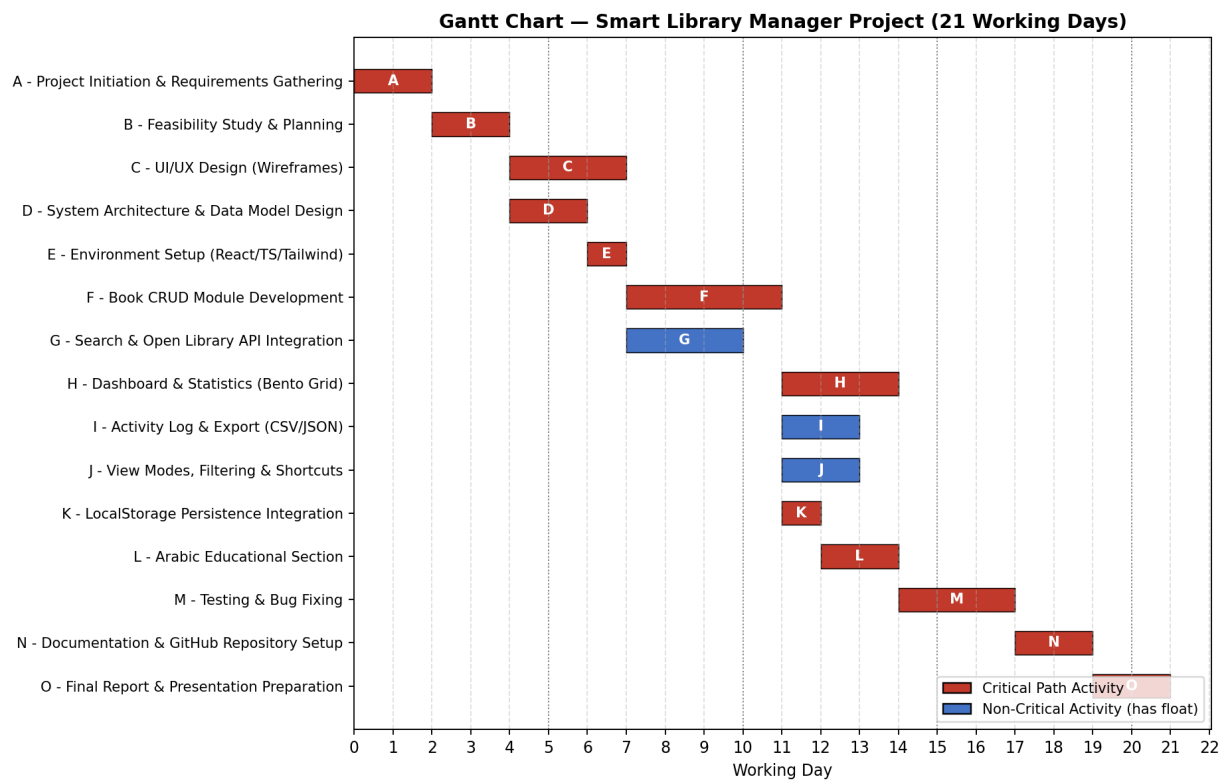
Using the normal distribution, the probability of completing the project within 23 working days (allowing a 1-2 day schedule buffer) is estimated as follows:

$$Z = (23 - 21.85) / 1.26 \approx 0.91$$

From the standard normal distribution table, $P(Z < 0.91) \approx 0.82$, meaning there is approximately an 82% probability that the project will be completed within 23 working days.

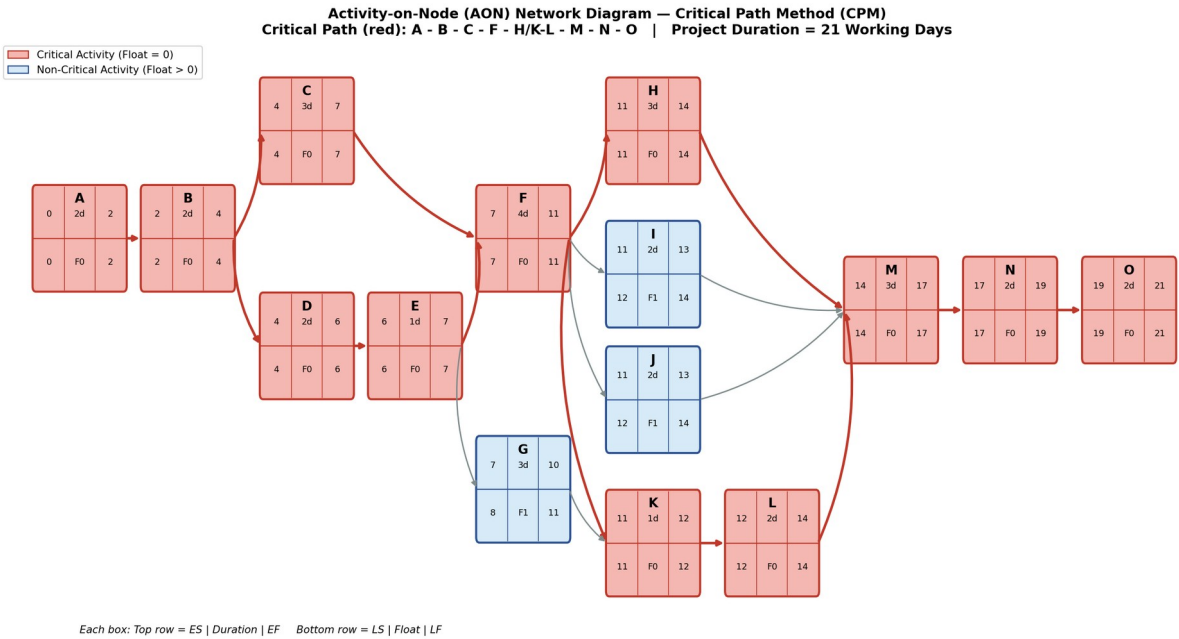
9. Gantt Chart

The Gantt chart below visualizes the schedule of all 15 project activities across the 21 working-day timeline, based on the Activity List and CPM analysis. Activities on the critical path are shown in red; activities with schedule float are shown in blue.



10. Network Diagram

The diagram below represents all project activities as nodes, connected according to their dependency (predecessor) relationships defined in the Activity List. Each node shows the Early Start (ES), Duration, Early Finish (EF), Late Start (LS), Total Float, and Late Finish (LF), calculated via forward and backward pass analysis.



11. Critical Path Method (CPM)

The Critical Path Method (CPM) identifies the longest sequence of dependent activities that determines the minimum possible project duration, and highlights which activities have zero scheduling flexibility (float).

2. Forward & Backward Pass Results

ID	ES	EF	LS	LF	Total Float	Critical?
A	0	2	0	2	0	Yes
B	2	4	2	4	0	Yes
C	4	7	4	7	0	Yes
D	4	6	4	6	0	Yes
E	6	7	6	7	0	Yes
F	7	11	7	11	0	Yes
G	7	10	8	11	1	No
H	11	14	11	14	0	Yes
I	11	13	12	14	1	No
J	11	13	12	14	1	No
K	11	12	11	12	0	Yes
L	12	14	12	14	0	Yes
M	14	17	14	17	0	Yes
N	17	19	17	19	0	Yes
O	19	21	19	21	0	Yes

3. Critical Path

Critical Path: A → B → C → F → H → M → N → O

An equally critical parallel path also exists: A → B → D → E → F → K → L → M → N → O (both paths share activities F, M, N, O and total exactly 21 working days).

Total Project Duration = 21 working days.

Activities G, I, and J each have 1 day of total float, meaning they can be delayed by up to 1 day without affecting the overall project completion date. All other activities are critical and have zero float — any delay in them directly delays the entire project.

4. Network Diagram Reference

See the separate Network Diagram document for the full Activity-on-Node visualization with the critical path highlighted in red.

12. Risk Analysis

The table below identifies the key risks facing the Smart Library Manager project, their likelihood and impact, and the planned mitigation strategy for each.

ID	Risk Description	Probability	Impact	Mitigation Strategy
R1	Open Library API downtime or rate-limiting	Medium	Medium	Cache recent results locally; allow manual book entry as fallback
R2	Data loss if browser cache/LocalStorage is cleared	Medium	High	Provide CSV/JSON export as a manual backup; remind users to export regularly
R3	Team member unavailability (exams, illness)	Medium	Medium	Cross-train members on tasks; build schedule buffer (float) into non-critical tasks
R4	Scope creep (adding features beyond agreed scope)	Medium	Medium	Enforce the Project Scope Statement; require team agreement for any scope change
R5	Browser compatibility issues (rendering/CSS)	Low	Medium	Test on multiple major browsers (Chrome, Edge, Firefox) before final delivery
R6	Integration bugs between search and CRUD modules	Medium	Medium	Perform incremental integration testing after each module is completed
R7	Underestimated learning curve for TypeScript	Medium	High	Start with simple typed components early; rely on official docs and tutorials
R8	GitHub repository not public or link broken at deadline	Low	High	Set up and verify the public repository early, well before the deadline

2. Risk Priority Summary

- High Impact Risks requiring closest monitoring: R2 (data loss), R7 (learning curve), R8 (repository access).
- Medium priority risks: R1, R3, R4, R6 — manageable with standard mitigation and monitoring.
- Low priority risk: R5 (browser compatibility) — addressed through routine testing.

3. Overall Risk Conclusion

No identified risk is severe enough to threaten the overall feasibility of the project. With the mitigation strategies above in place, the project team is confident the Smart Library Manager can be delivered successfully within the planned schedule and scope.